

AHAMMAD SHAWKI

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EDUCATION

Bangladesh University of Engineering and Technology (BUET)

BSc in Computer Science and Engineering

Expected 2028

Dhaka, Bangladesh

Coursework: Data Structures and Algorithms I and II, Database, Software Engineering, Object Oriented Programming, Computer Architecture, Theory of Computation, Discrete Mathematics.

EXPERIENCE

Founder and Developer, SrotDev

May 2025 to Present

- Lead an engineering collective shipping production web and mobile applications across React, Next.js, Django, FastAPI and PostgreSQL.
- Architected CareForAll as seven independently deployable Django services behind an API gateway, with its own Prometheus and Grafana observability stack.
- Built Srabon, a gamified science learning platform with multilingual and accessibility support, as the team's flagship product.

President of Web and App Development, NDITC

Mar 2022 to Jun 2024

- Led a 30-person engineering team; delivered the rebuilt official website and the UI/UX for the club's native mobile application.
- Raised over 1,000,000 BDT in sponsorship across 100+ company negotiations for INIT, the club's national annual event.
- Taught web development fundamentals directly to more than 500 students as a workshop instructor.

Founder and Team Lead, The AS8 Organization

Feb 2021 to May 2022

- Built a developer community that reached more than 20,000 learners through free programmes, courses and technical content.
- Partnered with Codecademy, Hack Club and DeepLearning.AI; wrote technical articles published on freeCodeCamp News.

PROJECTS

HALFSPREAD, autonomous options trading agent

Python, Alpaca, GitHub Actions

- Prices execution cost before every order, after measuring bid-ask cost ranging from 0.52% to 33% of contract value depending on time to expiry.
- Every published figure re-derives from an append-only journal with no API key and no network; the build fails if a number stops reproducing.

Cassandra, autonomous spreadsheet repair

Google ADK, Gemini, Cloud Run, Firestore

- Agent fleet that detects, repairs and verifies spreadsheet defects by recalculating the workbook; a correction that moves the wrong cell is rejected automatically.
- Repaired five defects in 114 seconds unattended on a model reporting \$6.2M in operating income for a company that had lost \$1.7M.

DeepSIFT, auditable digital forensics

MCP, Python, Volatility 3, MITRE ATT&CK

- 148-tool Model Context Protocol layer that parses forensic binaries into structured JSON before the model reads them, with a signable chain of custody.
- Scored 4/4 against published ground truth on two cases, with zero hallucinations and 100% claim grounding.

CADENCE, voice-based Parkinson's screening

PyTorch, scikit-learn, openSMILE, SHAP

- Screens from a 30-second voice sample and explains every score; validated across three independent language corpora using cross-database testing and domain-adversarial training to avoid microphone bias.

AWARDS

- Grand Prize Winner**, FLIR App Challenge 2025 to 2026, for SolarSnap.
- Winner, Best Overall Project**, ML Empowerment Build Challenge 2.0, 2026, for CADENCE.
- Winner, Best ERNIE Multimodal Application**, ERNIE AI Developer Challenge by Baidu, 2026, for Doclyst.
- Winner, Honourable Mention**, LMA EDGE Hackathon, 2026, presented at the London finale, for Coven.
- Honorable Mention Award**, Notre Dame College, 2023, for combined academic and leadership standing.

TECHNICAL SKILLS

Languages: Python, C/C++, TypeScript, JavaScript, Java, SQL, Dart, HTML/CSS

Frameworks: React, Next.js, Django, Django REST Framework, FastAPI, Flask, Node.js, Flutter, JavaFX, Tailwind CSS

AI and Machine Learning: PyTorch, scikit-learn, Model Context Protocol, Google ADK, Retrieval Augmented Generation, Gemini, Claude, Pandas, NumPy, SHAP

Infrastructure: Docker, Google Cloud (Cloud Run, Pub/Sub, Firestore, Vertex AI), AWS (Bedrock, ECS Fargate, Lambda), PostgreSQL, MongoDB, Redis, GitHub Actions, Prometheus, Grafana